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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25VDE101 – DIGITAL VLSI DESIGN

(For VDE)

Duration: 3 Hours]

[Max. Marks: 100

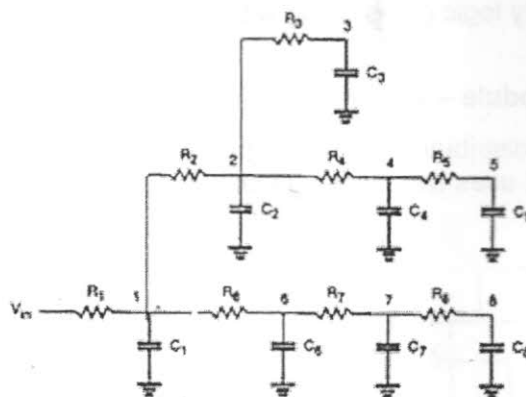
Note: Answer **Five full** questions choosing **One full** question from **each Module**.

Module – I

	Marks	BT*	CO*	PO*
1. a) With necessary derivations, calculate V_{OH} and V_{OL} in a resistive load inverter given that $V_{DD}=5$ V, $k_n=20 \mu A/V^2$, $V_{To}=0.8$ V, $R_L=200$ k Ω and $(W/L)_n=2$.	10	3	1	3
b) Outline the principles of constant-field scaling and analyze how scaling all device parameters by the same factor influences the gate-oxide capacitance, drain-current behavior in both the linear and saturation regions, and the resulting changes in total power dissipation and power density	10	2	1	3
2. a) For an NMOS inverter with a resistive load, derive the expression for the noise margin high (NM_H) and explain its significance in inverter operation.	10	2	1	3
b) Determine the type of MOSFET by computing the threshold voltage given that $t_{ox}=500$ Å, $\phi_{sc}=-0.94$ V, substrate doping concentration, $N_A=10^{16}$ cm $^{-3}$, $n_i=1.45 \times 10^{10}$ cm $^{-3}$, $Q_{ox}=3.2 \times 10^{-9}$ C/cm 2 , $\epsilon_{ox}=3.97\epsilon_0$ and $\epsilon_{Si}=11.7\epsilon_0$. Assume that $V_{SB}=0$.	10	3	1	3

Module – II

3. a) Describe the average current approach for determining the propagation delay T_{PHL} of a CMOS inverter. Using this method, compute the time it takes for the inverter output to drop from 90% to 10% of its high voltage level. Given: $V_{DD}=5$ V, $C_{out}=1$ pF, $\mu_n C_{ox}=20 \mu A/V^2$, $(W/L)_n=10$ and $V_{T,n}=1.0$ V.	10	3	2	3
b) Write the simple lumped RC representation, the T-model, and the distributed RC ladder model used for interconnects. For the RC tree network shown in Figure, determine the Elmore delay at nodes 5 and 7, assuming every resistor has a value of 2 k Ω and every capacitor has a value of 1 fF.				



10 3 2 3



4. a) Using the appropriate voltage and current waveforms, develop the mathematical expressions that describe power dissipation in a CMOS inverter. Briefly comment on the obtained expression. Also calculate the dynamic power dissipated by a CMOS inverter given a load capacitance of 20 fF, supply voltage of 1.2 V and an operating frequency of 200 MHz.
- b) Calculate noise margin NM_L in a CMOS inverter with the following parameters: $V_{DD} = 3.3$ V, $V_{T0,n} = 0.6$ V, $V_{T0,p} = -0.7$ V, $k_n = 200 \mu A/V^2$ and $k_p = 80 \mu A/V^2$. Derive necessary equations

10	3	2	3
10	3	2	3

Module – III

5. a) Describe how the bootstrapping method is used to eliminate the effective threshold-voltage limitation in NMOS-based circuits.
- b) Implement the logic function, $Y = A1A2 + (A3 + A4)(A5 + A6) + A7A8$ using static CMOS logic, dynamic CMOS and Domino CMOS logic.
6. a) i) Describe how data is transferred in a Enhancement-load, ratio less NMOS dynamic shift register that operates with a two-phase clocking scheme.
- ii) Explain the data-shifting process in a dynamic shift register built using single-phase CMOS transmission gates.
- b) Explain the concept of multiple output domino logic. Also design a multiple output domino CMOS gate to realize four outputs C_1, C_2, C_3 and C_4 defined by,

10	2	3	3
10	3	3	3
10	2	3	3

$$C_1 = G_1 + P_1 C_0 ; \quad C_2 = G_2 + P_2 G_1 + P_2 P_1 C_0$$

$$C_3 = G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 C_0$$

$$C_4 = G_4 + P_4 G_3 + P_4 P_3 G_2 + P_4 P_3 P_2 G_1 + P_4 P_3 P_2 P_1 C_0$$

10	3	3	3
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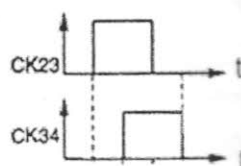
Module – IV

7. a) Draw a 4×4 barrel shifter using transmission gates, and explain how it performs both arithmetic and logical shift operations.
- b) i) Construct adder/subtractor using XOR gates and adders
- ii) Draw the logic diagram for zero and one detectors.
8. a) Design a single bit full adder using minimum number of MOSFETs and draw the CMOS circuit for the same. Also draw the symmetric topology.
- b) For two 4-bit inputs, X ($X_0 X_1 X_2 X_3$) and Y ($Y_0 Y_1 Y_2 Y_3$), explain the array multiplication process. Then, illustrate the block diagram of a 4×4 array multiplier built using the necessary logic gates along with 1-bit full adders and half adders.

10	2	4	3
10	3	4	3
10	3	4	3
10	2	4	3

Module – V

9. a) Explain how the H-tree method distributes a clock signal with matched delays. Also, design a circuit that uses two clocks as shown in Figure to generate four clock phases.



10	3	5	3
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10.	b)	Using appropriate illustrations, describe how the inherent parasitic bipolar transistors within CMOS fabrication can interact to trigger latch-up. Also list the techniques to prevent this phenomenon.	10	2	5	3
	a)	Explain why DFM is important, and use diagrams to show how process and design parameters relate and impact circuit/system performance.	10	2	5	3
	b)	Explain two types of output circuits employed on a chip. Also, design a level shifter that translates 3.3 V logic to 5 V logic, assuming $ V_{T0} = 1.0$ V and $\mu_p/\mu_n = 3$.	10	3	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
